



Inferring Security-relevant Properties from Java Documentation

Connor Myers

45323979

Supervised by

Mark Utting, Behnaz Hassanshahi, Dai Nguyen

School of Information Technology and Electrical Engineering
University of Queensland

Submitted for the degree of Bachelor of Engineering (Honours)
in the division of Software Engineering

July 29, 2022

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vivamus nulla libero, semper id dui finibus, suscipit pretium purus. Duis vehicula eleifend justo, at cursus turpis vulputate eget. Nullam vestibulum id neque a convallis. Sed molestie vehicula turpis, vel aliquet felis efficitur a. Fusce accumsan metus augue, id sagittis eros elementum vitae. Cras vel ultrices odio. Sed ac augue ex. Etiam tristique dignissim nibh, sit amet tincidunt libero tempor sit amet. Suspendisse tincidunt vitae ligula a posuere. Morbi euismod felis quis risus malesuada cursus. Curabitur imperdiet pretium odio, ac posuere turpis vehicula in. Praesent venenatis urna in augue congue elementum. Nam ligula nibh, rhoncus vitae luctus vel, feugiat sed neque. Curabitur ut arcu sed magna vulputate sollicitudin id a libero. Praesent tempus leo vel eros ornare mollis. Sed suscipit at mi ut viverra.

Declaration

This thesis is composed of my original work, and contains no material previously published or written by another person except where due reference has been made in the text. I have clearly stated the contribution by others to jointly-authored works that I have included in my thesis.

I have clearly stated the contribution of others to my thesis as a whole, including statistical assistance, survey design, data analysis, significant technical procedures, professional editorial advice, financial support and any other original research work used or reported in my thesis. The content of my thesis is the result of work I have carried out since the commencement of my higher degree by research candidature and does not include a substantial part of work that has been submitted to qualify for the award of any other degree or diploma in any university or other tertiary institution. I have clearly stated which parts of my thesis, if any, have been submitted to qualify for another award.

I acknowledge that an electronic copy of my thesis must be lodged with the University Library and, subject to the policy and procedures of The University of Queensland, the thesis be made available for research and study in accordance with the Copyright Act 1968 unless a period of embargo has been approved by the Dean of the Graduate School.

I acknowledge that copyright of all material contained in my thesis resides with the copyright holder(s) of that material. Where appropriate I have obtained copyright permission from the copyright holder to reproduce material in this thesis and have sought permission from co-authors for any jointly authored works included in the thesis.

Acknowledgements

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vivamus nulla libero, semper id dui finibus, suscipit pretium purus. Duis vehicula eleifend justo, at cursus turpis vulputate eget. Nullam vestibulum id neque a convallis. Sed molestie vehicula turpis, vel aliquet felis efficitur a. Fusce accumsan metus augue, id sagittis eros elementum vitae. Cras vel ultrices odio. Sed ac augue ex. Etiam tristique dignissim nibh, sit amet tincidunt libero tempor sit amet. Suspendisse tincidunt vitae ligula a posuere. Morbi euismod felis quis risus malesuada cursus. Curabitur imperdiet pretium odio, ac posuere turpis vehicula in. Praesent venenatis urna in augue congue elementum. Nam ligula nibh, rhoncus vitae luctus vel, feugiat sed neque. Curabitur ut arcu sed magna vulputate sollicitudin id a libero. Praesent tempus leo vel eros ornare mollis. Sed suscipit at mi ut viverra.

Contents

1	Introduction	5
2	Background Knowledge	7
3	Literature Review	9
4	Milestone Task 1	11
5	Milestone Task 2	13
6	Conclusion	15
A	Appendix	17

Chapter 1

Introduction

Static application security testing (SAST) applies static program analysis to the security of software projects. Static analysis is performed on the source code, compiled byte-code, or some other representation of the source code [1]. Popular SAST tools include: SonarQube, SpotBugs, Snyk Code, and Facebook’s Infer. SAST tools have become widely used at all stages of the software development life-cycle. Furthermore, they have been demonstrated to be highly effective in identifying defects in code compared to other methods, such as manual inspection [2].

Unfortunately, SAST tools are not perfect. From Rice’s Theorem, it is known that any non-trivial property about programs is reducible to the halting problem, i.e., undecidable [3]. That is, a perfect analysis is not always possible, and approximations must be made in order for the analysis to terminate, resulting in an imperfect output. False positives (where safe code is incorrectly labelled as a defect) and false negatives (where the analysis misses a genuine fault) [4] are common in all static analysis tools. False positives are not ideal as manual inspection is required to isolate genuine defects, leading to an increased time and labour cost. And, false negatives of course mean that genuine software faults are left undetected by the analysis. While solving the halting problem may be out of scope for an undergraduate thesis, there are still additional methodologies for improving existing static analyses that do not involve upsetting Mr. Turing.

To reduce the rate of false negatives, manual configuration of the SAST tools for specific software projects is often required [5]. For example, to perform an effective taint analysis, a custom configuration of sources, sinks and sanitisers for a particular project is often required. These taint analysis tools are not smart enough to know if a method is a source, sink, or a sanitiser — they require sensitive methods to be pre-labelled to conduct the analysis. Generally, these tools come with a default configuration containing a list of sensitive methods in popular APIs. For example,

the `getParameter()` method as a part of the `ServletRequest` interface would likely be listed by all taint analysis tools as a source. Of course, if a project is using a niche library or code written in-house, the taint analysis will not be able to incorporate those data-points into its analysis, leading to a sub-optimal analysis. To improve the taint analysis, one can always identify the sensitive methods manually and add these to the taint analyses configuration. However, creating this manual configuration is often a time-consuming and resource-intensive process. As such, methodologies to automatically label security sensitive methods may enable existing taint analysis tools to be configured more efficiently and effectively.

Deep learning, a subtopic within the broader machine learning, has recently increased in popularity. What was once a niche topic reserved for academics with access to computational resources far out of reach by most has become arguably one of the most influential technologies of the modern age. Deep learning has fundamentally changed our everyday life, with applications ranging from targeted advertising to identifying COVID-19 cases from lung CT scans [6]. Natural-language processing (NLP) is one often overlooked application of deep learning, but it has seen tremendous growth in recent years.

NLP has existed in one form or another for decades. In the early days of NLP, work was painstakingly done using handwritten rules, before moving to probabilistic methods in the 1990s, and then to deep learning in the 2010s [7]. With tech conglomerates such as Google and Microsoft publicly releasing large pre-trained models, NLP has never been more accessible. While machine translation and speech recognition are among the most popular applications of NLP, one application that has not been explored thoroughly in the literature is software security.

While being an informal description of code, Javadoc (i.e., natural language descriptions of code) can contain information relevant to software security. Using powerful pre-trained models such as CodeBERT that understand the semantic connection between code and natural language [8], it may be possible to extract security-relevant properties from Javadocs and use this information to configure existing SAST tools.

Chapter 2

Background Knowledge

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vivamus nulla libero, semper id dui finibus, suscipit pretium purus. Duis vehicula eleifend justo, at cursus turpis vulputate eget. Nullam vestibulum id neque a convallis. Sed molestie vehicula turpis, vel aliquet felis efficitur a. Fusce accumsan metus augue, id sagittis eros elementum vitae. Cras vel ultrices odio. Sed ac augue ex. Etiam tristique dignissim nibh, sit amet tincidunt libero tempor sit amet. Suspendisse tincidunt vitae ligula a posuere. Morbi euismod felis quis risus malesuada cursus. Curabitur imperdiet pretium odio, ac posuere turpis vehicula in. Praesent venenatis urna in augue congue elementum. Nam ligula nibh, rhoncus vitae luctus vel, feugiat sed neque. Curabitur ut arcu sed magna vulputate sollicitudin id a libero. Praesent tempus leo vel eros ornare mollis. Sed suscipit at mi ut viverra.

Maecenas vitae tempus arcu. Nullam maximus id ligula a finibus. Aliquam laoreet faucibus sem. Mauris interdum erat vel pretium iaculis. Sed et pretium arcu, at condimentum lorem. Suspendisse tincidunt neque vel ullamcorper consequat. Phasellus eget ligula quis velit fringilla porttitor. Pellentesque quis risus id erat euismod bibendum mattis nec sapien. Duis gravida, turpis at dapibus consequat, ex mi tempor augue, in scelerisque mi nisi ut lorem. Donec orci tellus, lobortis ac pretium non, elementum et dolor. Morbi sed hendrerit erat. Praesent nec ante lorem. Vestibulum dapibus varius nisl at hendrerit. Praesent aliquam vehicula dolor et pulvinar. Aenean facilisis leo eget lacus rhoncus, ac tempus urna molestie. Vivamus efficitur erat quis felis pulvinar mattis quis volutpat lorem.

Nullam luctus velit varius massa suscipit fringilla. Quisque egestas diam nec eros sagittis posuere. Sed venenatis facilisis nibh ac mattis. Morbi a mi at est vehicula imperdiet. Fusce tempus justo at tortor volutpat vulputate. Vestibulum molestie congue interdum. Nullam auctor faucibus risus ac hendrerit. Nulla viverra fermentum vulputate. Nulla mattis elementum ligula. Fusce eu tellus sapien. Maecenas scelerisque odio lacus, quis venenatis sapien molestie semper. Donec in consequat libero, quis euismod tellus.

Vestibulum eget lobortis lorem, id aliquam leo. Morbi ornare, enim et iaculis luctus, erat dui rutrum ligula, vel aliquam libero nisi vitae eros. Proin sodales ipsum nec ipsum tristique, porttitor sagittis tortor rhoncus. Integer auctor rutrum imperdiet. Nulla sollicitudin facilisis mauris, in mattis lorem ultrices quis. Curabitur neque lectus, auctor at ullamcorper et, porta eu nulla. Cras efficitur felis in ipsum volutpat, vitae condimentum velit ornare.

Donec consectetur, libero tempus sollicitudin gravida, felis erat aliquam libero, viverra maximus lacus nisi in lacus. Aenean hendrerit volutpat felis, sed viverra ex accumsan a. Vivamus vel tortor viverra nisi tincidunt dapibus. Proin diam libero, varius at turpis et, tincidunt congue arcu. Nunc tristique elit libero, nec rutrum quam aliquet at. Orci varius natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Vestibulum at nisi quis nulla malesuada tempor. Mauris eu felis risus. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia curae; Donec convallis est ex, vitae facilisis eros commodo in. Nullam faucibus nunc ac magna congue, et porttitor magna eleifend. Nunc at velit ullamcorper, pellentesque lorem in, dictum arcu. Suspendisse nec auctor velit. Mauris congue magna molestie augue maximus, ac hendrerit ante viverra. Pellentesque convallis nisl et felis venenatis varius.

Chapter 3

Literature Review

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vivamus nulla libero, semper id dui finibus, suscipit pretium purus. Duis vehicula eleifend justo, at cursus turpis vulputate eget. Nullam vestibulum id neque a convallis. Sed molestie vehicula turpis, vel aliquet felis efficitur a. Fusce accumsan metus augue, id sagittis eros elementum vitae. Cras vel ultrices odio. Sed ac augue ex. Etiam tristique dignissim nibh, sit amet tincidunt libero tempor sit amet. Suspendisse tincidunt vitae ligula a posuere. Morbi euismod felis quis risus malesuada cursus. Curabitur imperdiet pretium odio, ac posuere turpis vehicula in. Praesent venenatis urna in augue congue elementum. Nam ligula nibh, rhoncus vitae luctus vel, feugiat sed neque. Curabitur ut arcu sed magna vulputate sollicitudin id a libero. Praesent tempus leo vel eros ornare mollis. Sed suscipit at mi ut viverra.

Maecenas vitae tempus arcu. Nullam maximus id ligula a finibus. Aliquam laoreet faucibus sem. Mauris interdum erat vel pretium iaculis. Sed et pretium arcu, at condimentum lorem. Suspendisse tincidunt neque vel ullamcorper consequat. Phasellus eget ligula quis velit fringilla porttitor. Pellentesque quis risus id erat euismod bibendum mattis nec sapien. Duis gravida, turpis at dapibus consequat, ex mi tempor augue, in scelerisque mi nisi ut lorem. Donec orci tellus, lobortis ac pretium non, elementum et dolor. Morbi sed hendrerit erat. Praesent nec ante lorem. Vestibulum dapibus varius nisl at hendrerit. Praesent aliquam vehicula dolor et pulvinar. Aenean facilisis leo eget lacus rhoncus, ac tempus urna molestie. Vivamus efficitur erat quis felis pulvinar mattis quis volutpat lorem.

Nullam luctus velit varius massa suscipit fringilla. Quisque egestas diam nec eros sagittis posuere. Sed venenatis facilisis nibh ac mattis. Morbi a mi at est vehicula imperdiet. Fusce tempus justo at tortor volutpat vulputate. Vestibulum molestie congue interdum. Nullam auctor faucibus risus ac hendrerit. Nulla viverra fermentum vulputate. Nulla mattis elementum ligula. Fusce eu tellus sapien. Maecenas scelerisque odio lacus, quis venenatis sapien molestie semper. Donec in consequat libero, quis euismod tellus.

Vestibulum eget lobortis lorem, id aliquam leo. Morbi ornare, enim et iaculis luctus, erat dui rutrum ligula, vel aliquam libero nisi vitae eros. Proin sodales ipsum nec ipsum tristique, porttitor sagittis tortor rhoncus. Integer auctor rutrum imperdiet. Nulla sollicitudin facilisis mauris, in mattis lorem ultrices quis. Curabitur neque lectus, auctor at ullamcorper et, porta eu nulla. Cras efficitur felis in ipsum volutpat, vitae condimentum velit ornare.

Donec consectetur, libero tempus sollicitudin gravida, felis erat aliquam libero, viverra maximus lacus nisi in lacus. Aenean hendrerit volutpat felis, sed viverra ex accumsan a. Vivamus vel tortor viverra nisi tincidunt dapibus. Proin diam libero, varius at turpis et, tincidunt congue arcu. Nunc tristique elit libero, nec rutrum quam aliquet at. Orci varius natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Vestibulum at nisi quis nulla malesuada tempor. Mauris eu felis risus. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia curae; Donec convallis est ex, vitae facilisis eros commodo in. Nullam faucibus nunc ac magna congue, et porttitor magna eleifend. Nunc at velit ullamcorper, pellentesque lorem in, dictum arcu. Suspendisse nec auctor velit. Mauris congue magna molestie augue maximus, ac hendrerit ante viverra. Pellentesque convallis nisl et felis venenatis varius.

Chapter 4

Milestone Task 1

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vivamus nulla libero, semper id dui finibus, suscipit pretium purus. Duis vehicula eleifend justo, at cursus turpis vulputate eget. Nullam vestibulum id neque a convallis. Sed molestie vehicula turpis, vel aliquet felis efficitur a. Fusce accumsan metus augue, id sagittis eros elementum vitae. Cras vel ultrices odio. Sed ac augue ex. Etiam tristique dignissim nibh, sit amet tincidunt libero tempor sit amet. Suspendisse tincidunt vitae ligula a posuere. Morbi euismod felis quis risus malesuada cursus. Curabitur imperdiet pretium odio, ac posuere turpis vehicula in. Praesent venenatis urna in augue congue elementum. Nam ligula nibh, rhoncus vitae luctus vel, feugiat sed neque. Curabitur ut arcu sed magna vulputate sollicitudin id a libero. Praesent tempus leo vel eros ornare mollis. Sed suscipit at mi ut viverra.

Maecenas vitae tempus arcu. Nullam maximus id ligula a finibus. Aliquam laoreet faucibus sem. Mauris interdum erat vel pretium iaculis. Sed et pretium arcu, at condimentum lorem. Suspendisse tincidunt neque vel ullamcorper consequat. Phasellus eget ligula quis velit fringilla porttitor. Pellentesque quis risus id erat euismod bibendum mattis nec sapien. Duis gravida, turpis at dapibus consequat, ex mi tempor augue, in scelerisque mi nisi ut lorem. Donec orci tellus, lobortis ac pretium non, elementum et dolor. Morbi sed hendrerit erat. Praesent nec ante lorem. Vestibulum dapibus varius nisl at hendrerit. Praesent aliquam vehicula dolor et pulvinar. Aenean facilisis leo eget lacus rhoncus, ac tempus urna molestie. Vivamus efficitur erat quis felis pulvinar mattis quis volutpat lorem.

Nullam luctus velit varius massa suscipit fringilla. Quisque egestas diam nec eros sagittis posuere. Sed venenatis facilisis nibh ac mattis. Morbi a mi at est vehicula imperdiet. Fusce tempus justo at tortor volutpat vulputate. Vestibulum molestie congue interdum. Nullam auctor faucibus risus ac hendrerit. Nulla viverra fermentum vulputate. Nulla mattis elementum ligula. Fusce eu tellus sapien. Maecenas scelerisque odio lacus, quis venenatis sapien molestie semper. Donec in consequat libero, quis euismod tellus.

Vestibulum eget lobortis lorem, id aliquam leo. Morbi ornare, enim et iaculis luctus, erat dui rutrum ligula, vel aliquam libero nisi vitae eros. Proin sodales ipsum nec ipsum tristique, porttitor sagittis tortor rhoncus. Integer auctor rutrum imperdiet. Nulla sollicitudin facilisis mauris, in mattis lorem ultrices quis. Curabitur neque lectus, auctor at ullamcorper et, porta eu nulla. Cras efficitur felis in ipsum volutpat, vitae condimentum velit ornare.

Donec consectetur, libero tempus sollicitudin gravida, felis erat aliquam libero, viverra maximus lacus nisi in lacus. Aenean hendrerit volutpat felis, sed viverra ex accumsan a. Vivamus vel tortor viverra nisi tincidunt dapibus. Proin diam libero, varius at turpis et, tincidunt congue arcu. Nunc tristique elit libero, nec rutrum quam aliquet at. Orci varius natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Vestibulum at nisi quis nulla malesuada tempor. Mauris eu felis risus. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia curae; Donec convallis est ex, vitae facilisis eros commodo in. Nullam faucibus nunc ac magna congue, et porttitor magna eleifend. Nunc at velit ullamcorper, pellentesque lorem in, dictum arcu. Suspendisse nec auctor velit. Mauris congue magna molestie augue maximus, ac hendrerit ante viverra. Pellentesque convallis nisl et felis venenatis varius.

Chapter 5

Milestone Task 2

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vivamus nulla libero, semper id dui finibus, suscipit pretium purus. Duis vehicula eleifend justo, at cursus turpis vulputate eget. Nullam vestibulum id neque a convallis. Sed molestie vehicula turpis, vel aliquet felis efficitur a. Fusce accumsan metus augue, id sagittis eros elementum vitae. Cras vel ultrices odio. Sed ac augue ex. Etiam tristique dignissim nibh, sit amet tincidunt libero tempor sit amet. Suspendisse tincidunt vitae ligula a posuere. Morbi euismod felis quis risus malesuada cursus. Curabitur imperdiet pretium odio, ac posuere turpis vehicula in. Praesent venenatis urna in augue congue elementum. Nam ligula nibh, rhoncus vitae luctus vel, feugiat sed neque. Curabitur ut arcu sed magna vulputate sollicitudin id a libero. Praesent tempus leo vel eros ornare mollis. Sed suscipit at mi ut viverra.

Maecenas vitae tempus arcu. Nullam maximus id ligula a finibus. Aliquam laoreet faucibus sem. Mauris interdum erat vel pretium iaculis. Sed et pretium arcu, at condimentum lorem. Suspendisse tincidunt neque vel ullamcorper consequat. Phasellus eget ligula quis velit fringilla porttitor. Pellentesque quis risus id erat euismod bibendum mattis nec sapien. Duis gravida, turpis at dapibus consequat, ex mi tempor augue, in scelerisque mi nisi ut lorem. Donec orci tellus, lobortis ac pretium non, elementum et dolor. Morbi sed hendrerit erat. Praesent nec ante lorem. Vestibulum dapibus varius nisl at hendrerit. Praesent aliquam vehicula dolor et pulvinar. Aenean facilisis leo eget lacus rhoncus, ac tempus urna molestie. Vivamus efficitur erat quis felis pulvinar mattis quis volutpat lorem.

Nullam luctus velit varius massa suscipit fringilla. Quisque egestas diam nec eros sagittis posuere. Sed venenatis facilisis nibh ac mattis. Morbi a mi at est vehicula imperdiet. Fusce tempus justo at tortor volutpat vulputate. Vestibulum molestie congue interdum. Nullam auctor faucibus risus ac hendrerit. Nulla viverra fermentum vulputate. Nulla mattis elementum ligula. Fusce eu tellus sapien. Maecenas scelerisque odio lacus, quis venenatis sapien molestie semper. Donec in consequat libero, quis euismod tellus.

Vestibulum eget lobortis lorem, id aliquam leo. Morbi ornare, enim et iaculis luctus, erat dui rutrum ligula, vel aliquam libero nisi vitae eros. Proin sodales ipsum nec ipsum tristique, porttitor sagittis tortor rhoncus. Integer auctor rutrum imperdiet. Nulla sollicitudin facilisis mauris, in mattis lorem ultrices quis. Curabitur neque lectus, auctor at ullamcorper et, porta eu nulla. Cras efficitur felis in ipsum volutpat, vitae condimentum velit ornare.

Donec consectetur, libero tempus sollicitudin gravida, felis erat aliquam libero, viverra maximus lacus nisi in lacus. Aenean hendrerit volutpat felis, sed viverra ex accumsan a. Vivamus vel tortor viverra nisi tincidunt dapibus. Proin diam libero, varius at turpis et, tincidunt congue arcu. Nunc tristique elit libero, nec rutrum quam aliquet at. Orci varius natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Vestibulum at nisi quis nulla malesuada tempor. Mauris eu felis risus. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia curae; Donec convallis est ex, vitae facilisis eros commodo in. Nullam faucibus nunc ac magna congue, et porttitor magna eleifend. Nunc at velit ullamcorper, pellentesque lorem in, dictum arcu. Suspendisse nec auctor velit. Mauris congue magna molestie augue maximus, ac hendrerit ante viverra. Pellentesque convallis nisl et felis venenatis varius.

Chapter 6

Conclusion

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vivamus nulla libero, semper id dui finibus, suscipit pretium purus. Duis vehicula eleifend justo, at cursus turpis vulputate eget. Nullam vestibulum id neque a convallis. Sed molestie vehicula turpis, vel aliquet felis efficitur a. Fusce accumsan metus augue, id sagittis eros elementum vitae. Cras vel ultrices odio. Sed ac augue ex. Etiam tristique dignissim nibh, sit amet tincidunt libero tempor sit amet. Suspendisse tincidunt vitae ligula a posuere. Morbi euismod felis quis risus malesuada cursus. Curabitur imperdiet pretium odio, ac posuere turpis vehicula in. Praesent venenatis urna in augue congue elementum. Nam ligula nibh, rhoncus vitae luctus vel, feugiat sed neque. Curabitur ut arcu sed magna vulputate sollicitudin id a libero. Praesent tempus leo vel eros ornare mollis. Sed suscipit at mi ut viverra.

Maecenas vitae tempus arcu. Nullam maximus id ligula a finibus. Aliquam laoreet faucibus sem. Mauris interdum erat vel pretium iaculis. Sed et pretium arcu, at condimentum lorem. Suspendisse tincidunt neque vel ullamcorper consequat. Phasellus eget ligula quis velit fringilla porttitor. Pellentesque quis risus id erat euismod bibendum mattis nec sapien. Duis gravida, turpis at dapibus consequat, ex mi tempor augue, in scelerisque mi nisi ut lorem. Donec orci tellus, lobortis ac pretium non, elementum et dolor. Morbi sed hendrerit erat. Praesent nec ante lorem. Vestibulum dapibus varius nisl at hendrerit. Praesent aliquam vehicula dolor et pulvinar. Aenean facilisis leo eget lacus rhoncus, ac tempus urna molestie. Vivamus efficitur erat quis felis pulvinar mattis quis volutpat lorem.

Nullam luctus velit varius massa suscipit fringilla. Quisque egestas diam nec eros sagittis posuere. Sed venenatis facilisis nibh ac mattis. Morbi a mi at est vehicula imperdiet. Fusce tempus justo at tortor volutpat vulputate. Vestibulum molestie congue interdum. Nullam auctor faucibus risus ac hendrerit. Nulla viverra fermentum vulputate. Nulla mattis elementum ligula. Fusce eu tellus sapien. Maecenas scelerisque odio lacus, quis venenatis sapien molestie semper. Donec in consequat libero, quis euismod tellus.

Vestibulum eget lobortis lorem, id aliquam leo. Morbi ornare, enim et iaculis luctus, erat dui rutrum ligula, vel aliquam libero nisi vitae eros. Proin sodales ipsum nec ipsum tristique, porttitor sagittis tortor rhoncus. Integer auctor rutrum imperdiet. Nulla sollicitudin facilisis mauris, in mattis lorem ultrices quis. Curabitur neque lectus, auctor at ullamcorper et, porta eu nulla. Cras efficitur felis in ipsum volutpat, vitae condimentum velit ornare.

Donec consectetur, libero tempus sollicitudin gravida, felis erat aliquam libero, viverra maximus lacus nisi in lacus. Aenean hendrerit volutpat felis, sed viverra ex accumsan a. Vivamus vel tortor viverra nisi tincidunt dapibus. Proin diam libero, varius at turpis et, tincidunt congue arcu. Nunc tristique elit libero, nec rutrum quam aliquet at. Orci varius natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Vestibulum at nisi quis nulla malesuada tempor. Mauris eu felis risus. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia curae; Donec convallis est ex, vitae facilisis eros commodo in. Nullam faucibus nunc ac magna congue, et porttitor magna eleifend. Nunc at velit ullamcorper, pellentesque lorem in, dictum arcu. Suspendisse nec auctor velit. Mauris congue magna molestie augue maximus, ac hendrerit ante viverra. Pellentesque convallis nisl et felis venenatis varius.

Appendix A

Appendix

Bibliography

- [1] Flemming Nielson, Hanne Nielson, and Chris Hankin. *Principles of Program Analysis*. Jan. 1999. ISBN: 978-3-642-08474-4. DOI: 10.1007/978-3-662-03811-6.
- [2] William Nichols. *The Cost and Benefits of Static Analysis During Development*. Mar. 2020.
- [3] Anders Møller and Michael I Schwartzbach. “Static program analysis”. In: *Notes*. Feb (2012).
- [4] Brian Chess and Gary McGraw. “Static analysis for security”. In: *IEEE security & privacy* 2.6 (2004), pp. 76–79.
- [5] Fiorella Zampetti et al. “How Open Source Projects Use Static Code Analysis Tools in Continuous Integration Pipelines”. In: May 2017, pp. 334–344. DOI: 10.1109/MSR.2017.2.
- [6] Mohammad Rahimzadeh, Abolfazl Attar, and Seyed Sakhaei. *A Fully Automated Deep Learning-based Network For Detecting COVID-19 from a New And Large Lung CT Scan Dataset*. June 2020. DOI: 10.1101/2020.06.08.20121541.
- [7] Rachana Patel and Sanskruti Patel. “Deep Learning for Natural Language Processing”. In: (Jan. 2019).
- [8] Zhangyin Feng et al. “CodeBERT: A Pre-Trained Model for Programming and Natural Languages”. In: Jan. 2020, pp. 1536–1547. DOI: 10.18653/v1/2020.findings-emnlp.139.